

Explore Jharkhand – Full Stack Project Viva Guide

1. Project Overview Explore Jharkhand is a full-stack tourism platform built using Node.js, Express, MongoDB, and a dynamic EJS-based frontend. It supports three roles: Admin Traveler Tourist Guide Each role has its own dashboard, permissions, and data-driven interface.
2. Folder & File Structure Main folders: host/frontend: HTML, CSS, JS, Bootstrap, EJS views backend: server.js, routes, controllers, models, middleware server.js is the main backend file connecting all logic, middleware, database, and frontend static files.
3. Frontend Details Pages: index.html, login/signup, dashboard, admin.ejs, guide.ejs CSS & JS are inside /assets Responsive design with Bootstrap Uses EJS to inject dynamic content from backend JavaScript handles button clicks, AJAX calls, dynamic rendering Image Path Tip: Always use /assets/images/... to prevent broken paths after deployment.
4. Backend – Node & Express Express.js powers routing and server logic server.js uses express.static() to serve frontend APIs are in /routes/ folder: /api/user /api/admin /api/guide Logic is separated into controllers/ Data models are in models/
5. Middleware Flow Custom middleware: isLoggedIn – verifies user session isAdmin – restricts route to admins isGuide – restricts guide access Example: js Copy Edit router.get("/dashboard", isLoggedIn, isAdmin, adminDashboardController);
6. MongoDB & Mongoose Uses MongoDB Atlas for cloud storage Data schemas: User, Place, Booking, Feedback Mongoose defines schema and handles queries js Copy Edit const userSchema = new mongoose.Schema({ name: String, email: String, password: String, role: String, });
7. Role-Based Access Admin: Manage users, places, and guide approvals Traveler: View and book tourist places Guide: View booking requests and respond Middleware protects routes based on roles.
8. Deployment on Render Hosted at: <https://explore-jharkhand.onrender.com> Uses: js Copy Edit app.listen(process.env.PORT || 10000, "0.0.0.0"); express.static() serves frontend Always use absolute paths for assets
9. Common Errors & Fixes
  - ✖ 502 Gateway Error: Happens when the Render server sleeps or crashes. Fix by: Binding to 0.0.0.0 Using correct static file paths
  - ✖ Image not loading: Path must begin with /assets/... not ./ or images/
  - ✖ Session not working: Missing session middleware or cookie-parser
10. Viva Q&A Quick Answers Q: Why use Express.js? A: Fast routing, easy middleware, and RESTful API support. Q: What is middleware? A: Function that processes request before final handler. Q: Why MongoDB? A: No-SQL, flexible for tourism data (places, feedback). Q: What is EJS? A: Embedded JavaScript to inject backend variables into HTML. Q: How is frontend connected? A: Via routes (e.g. form submission, fetch API).
11. Final Notes Explore Jharkhand is a modular, secure, and role-based full stack web app with: Node.js backend MongoDB database EJS + Bootstrap frontend Proper authentication and deployment ✖ You Can Add: Diagrams of folder structure Architecture flow: User ✖ API ✖ DB ✖ Response Screenshot of your dashboards